The Anthropic Gap

A Supplementary Learning Track

AI Safety · Mechanistic Interpretability · RLHF · Constitutional AI

The four gaps between a strong AI/ML engineer and Anthropic-level readiness.
Every resource is free. All links are verified.

What Is The Anthropic Gap?

Your core AI/ML Engineering Roadmap (Phases 0–7) will make you a strong, hireable AI/ML engineer at most companies. But Anthropic is not most companies. Their entire mission — and the lens through which they hire — is AI safety. This document covers the four gaps between a strong general AI/ML engineer and someone Anthropic will seriously consider. Every resource listed here is free. Complete these alongside and after Phase 5 of your main roadmap.

> **Important:** Anthropic interviews probe AI safety and interpretability directly. You can be the most brilliant engineer in the world, but if you cannot articulate a thoughtful perspective on the risks and challenges of advanced AI, you will not get far. These resources fix that gap.

The Four Gaps At a Glance

GAP 1	AI Safety and Alignment — Anthropic's hiring filter
GAP 2	Mechanistic Interpretability — the field Anthropic invented
GAP 3	RLHF and Constitutional AI — how Claude is actually built
GAP 4	Certification and Community — credibility and connections

GAP 1 — AI Safety and Alignment

Anthropic screens for mission alignment from the very first recruiter conversation. You need to understand why AI safety matters, what the open problems are, and how researchers approach them technically. This is not optional background reading — it is the filter that removes candidates before any technical interview.

STEP 1 — AI Safety Fundamentals — Alignment Course

Provider: BlueDot Impact (AI Safety Fundamentals) | **Level:** Beginner → Advanced | 8 weeks

Link: https://course.aisafetyfundamentals.com/alignment

What you will learn:

- Core concepts in AI safety and alignment from first principles
- Technical vs governance approaches to AI risk
- Inner alignment, outer alignment, and deceptive alignment
- Scalable oversight and debate as safety techniques
- Reward hacking, specification gaming, and goal misgeneralisation
- AI risk arguments: instrumental convergence, orthogonality thesis
- Current alignment research landscape and open problems
- Community of other safety-focused learners included

Why this resource: The most structured beginner-to-advanced alignment course available. Specifically designed for people who want to work in AI safety. Start here — it frames everything else in this gap.

STEP 2 — Stanford CS120 — Introduction to AI Safety

Provider: Stanford University (free lecture materials) | **Level:** Intermediate → Advanced | Full semester

Link: https://web.stanford.edu/class/cs120/

What you will learn:

- Technical challenges of creating reliable and aligned AI systems
- Model-specific vs systemic safety issues
- Fairness, data limitations, and adversarial vulnerabilities
- Embedding desired behaviour in AI systems
- Interpretability and robustness across RL, CV, and NLP
- Evaluating and auditing AI systems
- Sociotechnical concerns of modern AI deployment
- Future risks and oversight of intelligence

Why this resource: Stanford-level rigour on the technical side of AI safety. Goes deeper than BlueDot into the research methods. Do this after completing the BlueDot course — it will make far more sense with that foundation.

STEP 3 Anthropic Research Papers — Primary Source

Provider: Anthropic (free, publicly published) | **Level:** Intermediate → Advanced | Self-paced

Link: https://www.anthropic.com/research

What you will learn:

- Concrete Problems in AI Safety (Amodei et al., 2016) — read first
- Constitutional AI: Harmlessness from AI Feedback (2022)
- Scaling Laws for Neural Language Models (2020)
- Claude's Model Spec — the philosophical foundation of Claude's training
- Scaling Monosemanticity: Extracting Interpretable Features from Claude 3
- Alignment Science team publications on scalable oversight
- Red-teaming language models with language models

Why this resource: Anthropic interviewers will reference these papers directly. Reading them is not optional — it is the minimum expected preparation. Understand each paper well enough to explain it, critique it, and discuss what problems remain unsolved.

Additional Reading: The Alignment Forum (alignmentforum.org) and LessWrong (lesswrong.com) are where active safety researchers publish work-in-progress ideas. Reading these regularly builds the vocabulary and intuition that Anthropic expects.

GAP 2 — Mechanistic Interpretability

Anthropic's interpretability team — led by Chris Olah — essentially invented this field. The goal: reverse-engineer neural networks the way a programmer reverse-engineers compiled binary code, to understand what algorithms they are running. This is one of Anthropic's most active hiring areas. Your main roadmap does not touch this at all.

STEP 1 — Prerequisites Check — Neel Nanda's Barebones Guide

Provider: Neel Nanda (former Anthropic, ex-DeepMind interpretability team) | **Level:** Beginner → Intermediate | Read before anything else

Link: https://www.neelnanda.io/mechanistic-interpretability/prereqs

What you will learn:

- Exact skills required before starting mech interp — no more, no less
- Linear algebra: eigenvalues, eigenvectors, orthonormal bases
- Backpropagation: chain rule on multivariate functions
- Transformer architecture: deep understanding, not just usage
- Coding exercise: implement GPT-2 from scratch (with tests)
- Python proficiency at research level
- What to ignore — Neel is clear about what does NOT matter yet

Why this resource: Written by the person who built the field's main educational materials after working directly with Chris Olah at Anthropic. He tells you exactly what you need without overstating it. Read this first to know where you stand.

STEP 2 — ARENA — AI Safety and Mechanistic Interpretability

Provider: ARENA Education (Callum McDougall) | **Level:** Beginner → Advanced | Structured curriculum

Link: https://www.arena.education/chapter1

What you will learn:

- Build and train transformers from scratch with full code
- TransformerLens library for interpretability experiments
- Induction heads — what they are and why they matter
- Superposition: how models pack more features than dimensions
- Sparse Autoencoders (SAEs) — Anthropic's primary interpretability tool
- Linear probing — detecting what information models represent internally
- Activation steering — directly modifying model behaviour
- Circuits analysis: reverse-engineering the algorithms inside transformers
- Hands-on exercises with solutions throughout

Why this resource: The single best structured curriculum for mechanistic interpretability. Built specifically around Anthropic's Transformer Circuits research. Full of runnable notebooks and exercises. This is where you build the actual skills.

STEP 3 Transformer Circuits Thread — Anthropic's Research

Provider: Anthropic Interpretability Team (Chris Olah et al.) | **Level:** Intermediate → Advanced | Read progressively

Link: https://transformer-circuits.pub/

What you will learn:

- A Mathematical Framework for Transformer Circuits — the foundational paper
- In-Context Learning and Induction Heads — what induction heads do
- Toy Models of Superposition — how models store more features than dimensions
- Scaling Monosemanticity — finding interpretable features in Claude 3 Sonnet
- The Golden Gate Claude experiment — activating individual concepts
- 34 million features identified in Claude Sonnet via Sparse Autoencoders
- Ongoing circuit updates and latest team findings

Why this resource: This is the primary source — written by the team you would work alongside. Reading it is both preparation and demonstration of serious intent. Start with the Mathematical Framework, then read chronologically.

STEP 4 Neel Nanda's Getting Started Guide + YouTube Channel

Provider: Neel Nanda | **Level:** Beginner → Advanced | Ongoing reference

Link: https://www.neelnanda.io/mechanistic-interpretability/quickstart

What you will learn:

- Exact research mini-project ideas to build portfolio (1–5 day projects)
- Video walkthroughs of mech interp papers
- Implementing GPT-2 from scratch — video tutorial with exercises
- Introduction to TransformerLens via studying induction heads
- Research mindset: how to approach open problems
- How to go from learning to doing actual publishable research

Why this resource: The most practical roadmap for going from learning to doing research. Mech interp has short feedback loops and modest compute requirements — you can publish real findings. Neel explains exactly how to get there.

GAP 3 — RLHF and Constitutional AI

What is RLHF?
Reinforcement Learning from Human Feedback is the training technique that turns a raw language model into something safe and useful. Stage 1: supervised fine-tuning on good examples. Stage 2: humans compare model outputs and a reward model learns their preferences. Stage 3: the main model is trained via RL to score highly on the reward model. Constitutional AI (Anthropic's method) replaces human labellers in Stage 2 with AI feedback guided by a written constitution — this is how Claude is trained. DPO (Direct Preference Optimization) is a newer, simpler alternative that skips the reward model and optimises directly from preference data.

STEP 1 — Reinforcement Learning from Human Feedback — The RLHF Book

Provider: Nathan Lambert (Hugging Face RLHF researcher) | **Level:** Beginner → Advanced | Free web book

Link: https://rlhfbook.com

What you will learn:

- RLHF pipeline from scratch: SFT → reward model → PPO
- Reward model training and its failure modes (reward hacking)
- Direct Preference Optimization (DPO) — the simpler alternative to PPO
- Constitutional AI and AI Feedback (RLAIF) — Anthropic's training method
- Reasoning and Reinforcement Fine-Tuning (RLVR)
- Synthetic data: how models are now trained on AI-generated preferences
- Evaluation of aligned models
- Over-optimisation: why RLHF goes wrong

Why this resource: The canonical free reference on RLHF. Written because no comprehensive free resource existed. Covers Constitutional AI — Anthropic's specific training approach — in depth. Read this before attempting to implement anything.

STEP 2 — Foundational Papers — Read in This Order

Provider: OpenAI / Anthropic / Academic (all free) | **Level:** Intermediate → Advanced | Self-paced

Link: https://www.anthropic.com/research | arxiv.org

What you will learn:

- InstructGPT: Training Language Models to Follow Instructions (OpenAI, 2022)
- Constitutional AI: Harmlessness from AI Feedback (Anthropic, 2022)
- Direct Preference Optimization: Your Language Model is Secretly a Reward Model (2023)
- Scaling Laws for Neural Language Models (Kaplan et al., 2020)
- Deep Reinforcement Learning from Human Preferences (Christiano et al., 2017)
- Concrete Problems in AI Safety (Amodei et al., 2016)

Why this resource: Anthropic interviewers expect you to know these papers. Reading them is not optional. Understand the motivation, method, results, and limitations of each. Be ready to discuss what problems they solved and what they left open.

STEP 3

Hugging Face TRL Library — Hands-On RLHF Implementation

Provider: Hugging Face (official documentation, free) | **Level:** Intermediate → Advanced | Self-paced

Link: https://huggingface.co/docs/trl/index

What you will learn:

- SFTTrainer: supervised fine-tuning on your own data
- RewardTrainer: building and training reward models
- PPOTrainer: full RLHF training loop
- DPOTrainer: direct preference optimisation implementation
- KTO, ORPO, and other alignment training methods
- RLHF with LoRA and QLoRA for consumer hardware
- Working code examples for each training stage

Why this resource: This is where you move from understanding to building. Implement a reward model. Run a DPO training loop. These hands-on experiments are what you put on GitHub — and what Anthropic wants to see.

GAP 4 — Certification and Community

Anthropic hires people who are visibly part of the AI safety ecosystem — not just people who read about it privately. Certification gives you credibility on LinkedIn and your CV. Community gives you collaborators, feedback on your work, and occasionally referrals.

CERT — BlueDot Impact — AI Alignment Course with Certificate

Provider: BlueDot Impact | **Level:** Beginner → Advanced | Free with certificate

Link: https://bluedot.org/courses/alignment

What you will learn:

- Structured 8-week cohort format with other safety-focused learners
- Key concepts across technical AI safety and frontier AI governance
- Certificate of completion shareable on LinkedIn
- Access to a global community of people pursuing AI safety careers
- Fellowship and funding opportunities via the BlueDot network
- Introduction to MATS, ARC, Redwood Research and other safety orgs

Why this resource: This is the certification to get. It is specifically designed to build the knowledge and connections needed for a high-impact AI safety career. The community alone is worth the 8 weeks.

COMM — AI Safety Fundamentals Community + Alignment Forum

Provider: Multiple (all free) | **Level:** Beginner → Advanced | Ongoing

Link: https://alignmentforum.org | https://lesswrong.com | https://www.aisafety.com/self-study

What you will learn:

- Alignment Forum: where active safety researchers publish ideas
- LessWrong: broader rationalist and AI risk community
- AISafety.com self-study hub: curated reading lists and events
- MATS (ML Alignment Theory Scholars): mentored research programme
- Anthropic's own published research discussions
- Open problems list: real unsolved questions you can work on

Why this resource: Being part of this community signals genuine engagement. Posting thoughtful commentary on the Alignment Forum, or publishing a blog post about a mech interp experiment, builds visible credibility. This is the difference between understanding safety and being known in safety.

Exact Order — When to Do What

The main roadmap (Phases 0-7) stays unchanged. This track runs alongside and after it.
The table below shows exactly when to layer each gap resource in

Phase	When	Action
After Phase 5 (Deep Learning)	Run alongside Phase 6	Start the BlueDot AI Alignment course (8 weeks). This runs in parallel with your Phase 6 modern AI engineering work. BlueDot is self-paced and structured, so you can fit it alongside your technical study without falling behind on either track.
Phase 6.4 (Generative AI)	Simultaneously	Read the RLHF Book (rlhfbook.com) alongside your Generative AI course. The RLHF Book explains exactly how models like Claude are trained and aligned. Reading it here, while GenAI concepts are fresh, means the Constitutional AI and DPO sections will make immediate sense.
After Phase 6.4	1-2 weeks	Read the foundational papers in order: InstructGPT, then Constitutional AI, then DPO. Then implement a reward model and a DPO training loop using the Hugging Face TRL library. Push your implementation to GitHub. This is your first tangible Anthropic-relevant project.
After Phase 6 (complete)	2-3 weeks	Work through Stanford CS120: Introduction to AI Safety. Your Phase 6 technical depth makes this course land differently than it would have earlier. You now have the hands-on LLM and systems knowledge to engage with the safety material at a research level, not just conceptually.
After CS120	Ongoing	Read Neel Nanda's prerequisites guide to check where you stand. Work through ARENA Chapter 1 systematically, completing every exercise. Begin reading the Transformer Circuits thread in parallel, starting with the Mathematical Framework paper. Don't rush this: depth matters more than speed here.
Alongside ARENA	Ongoing	Read Anthropic's research papers as you encounter the relevant topics in ARENA. When ARENA covers induction heads, read the Induction Heads paper. When it covers superposition, read Toy Models of Superposition. Join the Alignment Forum and start reading regularly, even if you are not posting yet.
After ARENA	Ongoing	Do 1-5 day mech interp mini-projects using Neel Nanda's project list. Publish your findings on GitHub with a clear README. Write a short post on LinkedIn or the Alignment Forum explaining what you found. Even a small, honest experiment published publicly is worth more than any credential for getting noticed in the safety community.
Anytime	Apply when ready	Apply for the BlueDot certificate cohort when you are ready. Regularly read alignmentforum.org and lesswrong.com to stay current with the research conversation. Be visible in the safety community: comment, share, and engage with the work. Anthropic notices people who show up consistently.